# VIVEKANANDHA SAINIK COACHING CENTER

## ALL INDIA SAINIK SCHOOL ENTRANCE EXAMINATION (AISSEE) CLASS IX

**MOCK PAPER -** 08

**Time Allowed:** 3 Hours (180 Mins) | **Maximum Marks:** 400 | **Total Questions:** 150

**INSTRUCTIONS TO CANDIDATES:**
1. This paper contains 150 questions divided into 5 Sections: Section A (Mathematics - 50 Qs, 200 Marks), Section B (Intelligence - 25 Qs, 50 Marks), Section C (English - 25 Qs, 50 Marks), Section D (General Science - 25 Qs, 50 Marks), Section E (Social Science - 25 Qs, 50 Marks).
2. All questions are compulsory. There is no negative marking.
3. Use standard black/blue ballpoint pen to darken OMR bubbles.

# PART A: QUESTION PAPER

## SECTION A: MATHEMATICS

**Q1. The product of two rational numbers is -16/9. If one of the numbers is -4/3, find the other.**

(a) 4/3 (b) -4/3
(c) 3/4 (d) -3/4

**Q2. Solve for y: (y - 8)/3 = (y - 3)/2**

(a) y = -7 (b) y = 7
(c) y = -13 (d) y = 13

**Q3. A two-digit number is such that the sum of its digits is 11. If 27 is added to the number, its digits are reversed. Find the number.**

(a) 47 (b) 74
(c) 38 (d) 56

**Q4. Find the measure of each exterior angle of a regular polygon of 15 sides.**

(a) 24 degrees (b) 30 degrees
(c) 18 degrees (d) 36 degrees

**Q5. The adjacent sides of a rectangle are in the ratio 4 : 3 and its perimeter is 56 cm. Find the length of its diagonal.**

(a) 20 cm (b) 25 cm
(c) 15 cm (d) 18 cm

**Q6. A card is drawn at random from a deck of 52 cards. What is the probability of getting a red queen?**

(a) 1/26 (b) 1/13
(c) 1/52 (d) 2/13

**Q7. What is the least number that must be subtracted from 1300 so as to get a perfect square?**

(a) 4 (b) 9
(c) 12 (d) 16

**Q8. Find the square root of 0.00059049.**

(a) 0.0243 (b) 0.243
(c) 0.00243 (d) 0.043

**Q9. Is 68600 a perfect cube? If not, find the smallest number by which it must be multiplied to get a perfect cube.**

(a) 5 (b) 2
(c) 10 (d) 4

**Q10. What is the cube root of -2744?**

(a) -14 (b) 14
(c) -16 (d) 16

**Q11. An article was purchased for Rs 1239 including GST of 18%. Find the price of the article before GST was added.**

(a) Rs 1050 (b) Rs 1000
(c) Rs 1100 (d) Rs 1020

**Q12. Find the compound interest on Rs 12600 for 2 years at 10% per annum compounded annually.**

(a) Rs 2646 (b) Rs 2500
(c) Rs 2700 (d) Rs 2400

**Q13. Subtract 3xy + 5yz - 7zx from 5xy - 2yz - 2zx + 10xyz.**

(a) 2xy - 7yz + 5zx + 10xyz (b) 2xy + 3yz - 9zx + 10xyz
(c) 8xy - 3yz - 9zx (d) 2xy - 7yz - 9zx

**Q14. Using suitable identity, find (102)^2.**

(a) 10404 (b) 10400
(c) 10204 (d) 10604

**Q15. A solid has 12 edges and 6 faces. How many vertices does it have?**

(a) 8 (b) 6
(c) 10 (d) 12

**Q16. Find the area of a trapezium whose parallel sides are 24 cm and 20 cm and distance between them is 15 cm.**

(a) 330 cm^2 (b) 660 cm^2
(c) 300 cm^2 (d) 350 cm^2

**Q17. The volume of a cylinder is 1540 cm^3 and its height is 10 cm. Find the radius of its base. (Use pi = 22/7)**

(a) 7 cm (b) 14 cm
(c) 3.5 cm (d) 10 cm

**Q18. Find the volume of a cube whose total surface area is 486 cm^2.**

(a) 729 cm^3 (b) 243 cm^3
(c) 512 cm^3 (d) 343 cm^3

**Q19. Simplify: (3^-5 * 10^-5 * 125) / (5^-7 * 6^-5)**

(a) 3125 (b) 625
(c) 125 (d) 25

**Q20. If 15 men can complete a project in 24 days, how many men will complete the same project in 18 days?**

(a) 20 (b) 18
(c) 22 (d) 25

**Q21. Factorise: 16x^2 - 25y^2**

(a) (4x - 5y)(4x + 5y)  (b) (4x - 5y)^2
(c) (16x - 25y)(x + y)  (d) (2x - 5y)(8x + 5y)

**Q22. Factorise: p^2 + 6p + 8**

(a) (p + 2)(p + 4)  (b) (p - 2)(p - 4)
(c) (p + 1)(p + 8)  (d) (p + 3)(p + 3)

**Q23. If a 4-digit number 31z5 is a multiple of 3, where z is a digit, find the sum of all possible values of z.**

(a) 18  (b) 15
(c) 12  (d) 9

**Q24. Find the distance between points A(2, 0) and B(7, 0) on a line.**

(a) 5 units  (b) 9 units
(c) 7 units  (d) 2 units

**Q25. Simplify: (x^a / x^b)^(a+b) * (x^b / x^c)^(b+c) * (x^c / x^a)^(c+a)**

(a) 1  (b) x
(c) 0  (d) x^(a+b+c)

**Q26. The ratio of length to breadth of a rectangular field is 5 : 3. If the perimeter is 160 m, find its area.**

(a) 1500 m^2  (b) 1200 m^2
(c) 1600 m^2  (d) 1800 m^2

**Q27. A cylindrical well of depth 20 m and diameter 7 m is dug. Find the volume of earth taken out.**

(a) 770 m^3  (b) 385 m^3
(c) 1540 m^3  (d) 308 m^3

**Q28. Find the length of the longest rod that can be placed in a room 12 m long, 9 m broad and 8 m high.**

(a) 17 m  (b) 15 m
(c) 19 m  (d) 21 m

**Q29. Evaluate: [(1/3)^-1 - (1/4)^-1]^-1**

(a) -1  (b) 1
(c) 7  (d) -7

**Q30. A car travels 180 km in 3 hours. How far will it travel in 5 hours at the same speed?**

(a) 300 km  (b) 250 km
(c) 280 km  (d) 320 km

**Q31. Factorise: a^4 - b^4**

(a) (a - b)(a + b)(a^2 + b^2)  (b) (a - b)^2 (a + b)^2
(c) (a^2 - b^2)^2  (d) (a^3 - b^3)(a + b)

**Q32. Divide: 28x^4 by 56x**

(a) 1/2 x^3  (b) 2x^3
(c) 1/2 x^2  (d) x^3

**Q33. If 24x is a multiple of 6, where x is a digit, what are the possible values of x?**

(a) 0, 6  (b) 2, 8
(c) 0, 3, 6, 9  (d) 0, 2, 4, 6, 8

**Q34. In a frequency distribution table, the width of each class interval is 10 and the lower limit of the first class interval is 5. What is the upper limit of the fifth class interval?**

(a) 55 (b) 45
(c) 50 (d) 60

**Q35. The mean of 5 numbers is 27. If one number is excluded, their mean becomes 25. Find the excluded number.**

(a) 35 (b) 30
(c) 32 (d) 28

**Q36. Find the value of x if 2^(x - 3) = 16.**

(a) 7 (b) 6
(c) 5 (d) 4

**Q37. The measures of two angles of a triangle are 55 degrees and 65 degrees. Find the third angle.**

(a) 60 degrees (b) 50 degrees
(c) 70 degrees (d) 80 degrees

**Q38. Find the perimeter of a rhombus whose diagonals measure 16 cm and 12 cm.**

(a) 40 cm (b) 30 cm
(c) 50 cm (d) 48 cm

**Q39. A hemispherical bowl has a radius of 3.5 cm. Find its volume.**

(a) 89.83 cm^3 (b) 179.66 cm^3
(c) 90 cm^3 (d) 100 cm^3

**Q40. Simplify: (5^2)^3 * 5^-4**

(a) 25 (b) 5
(c) 125 (d) 1

**Q41. A wheel makes 360 revolutions in one minute. Through how many radians does it turn in one second?**

(a) 12 pi (b) 6 pi
(c) 24 pi (d) 18 pi

**Q42. Factorise: 100 - 9x^2**

(a) (10 - 3x)(10 + 3x) (b) (10 - 3x)^2
(c) (100 - 3x)(1 + 3x) (d) (10 + 3x)^2

**Q43. Divide: (z^2 - 4z - 12) by (z + 2)**

(a) z - 6 (b) z + 6
(c) z - 2 (d) z + 4

**Q44. If a coin is tossed 3 times, what is the total number of possible outcomes?**

(a) 8 (b) 6
(c) 4 (d) 16

**Q45. Find the square root of 529.**

(a) 23 (b) 27
(c) 17 (d) 29

**Q46. Find the smallest number by which 1250 must be divided to make it a perfect square.**

(a) 2 (b) 5
(c) 10 (d) 3

**Q47. A shopkeeper marks his goods 20% above CP and allows a discount of 10%. Find his profit percentage.**

(a) 8% (b) 10%
(c) 12% (d) 6%

**Q48. Express 4.5 * 10^-4 in usual form.**

(a) 0.00045 (b) 0.0045
(c) 0.045 (d) 0.000045

**Q49. If 10 men can reap a field in 8 days, how many days will 16 men take to reap the same field?**

(a) 5 days (b) 6 days
(c) 4 days (d) 8 days

**Q50. The sum of angles of a convex polygon of n sides is given by:**

(a) (n - 2) * 180 degrees (b) (n + 2) * 180 degrees
(c) n * 180 degrees (d) (2n - 1) * 90 degrees

## SECTION B: INTELLIGENCE

**Q51. If 'SAINIK' is coded as 'TBJOJL', how is 'SCHOOL' coded?**

(a) TDIPPM (b) TCHMNN
(c) TDJPPM (d) SEIPPM

**Q52. Find the next term in the series: 3, 7, 15, 31, 63, ?**

(a) 127 (b) 125
(c) 128 (d) 120

**Q53. Find the odd one out:**

(a) Square (b) Cube
(c) Rectangle (d) Triangle

**Q54. Pointing to a lady, a man said, 'Her mother's only daughter is my wife.' How is the lady related to the man?**

(a) Wife (b) Daughter
(c) Sister (d) Mother

**Q55. If '+' means '-', '-' means '*', '*' means '/' and '/' means '+', evaluate: 15 / 5 - 2 * 4 + 6**

(a) 11.5 (b) 12
(c) 10 (d) 15

**Q56. Complete the analogy: Eye : See :: Ear : ?**

(a) Hear (b) Speak
(c) Taste (d) Smell

**Q57. A boy walks 10 m West, turns Left and walks 10 m, turns Left and walks 10 m. How far is he from his starting point?**

(a) 10 m (b) 20 m
(c) 0 m (d) 15 m

**Q58. Arrange in a meaningful logical order: 1. Seed 2. Plant 3. Fruit 4. Tree 5. Flower**

(a) 1, 2, 4, 5, 3 (b) 1, 2, 3, 4, 5
(c) 2, 1, 4, 3, 5 (d) 1, 4, 2, 5, 3

**Q59. In a row of 30 girls, Radha is 10th from the left end. What is her position from the right end?**

(a) 21st (b) 20th
(c) 22nd (d) 19th

**Q60. Find the missing letter in series: B, D, G, K, P, ?**

(a) V (b) U
(c) W (d) T

**Q61. Select the word which CANNOT be formed from the letters of 'KNOWLEDGE':**

(a) DANGER (b) GOLDEN
(c) KNOW (d) LODGE

**Q62. Complete the analogy: 8 : 64 :: 11 : ?**

(a) 121 (b) 110
(c) 131 (d) 100

**Q63. If SUN = 54, then MOON = ?**

(a) 57 (b) 50
(c) 60 (d) 52

**Q64. Which diagram best represents: India, Delhi, Asia?**

(a) Three concentric circles (b) Three disjoint circles
(c) Two intersecting circles (d) One circle inside two disjoint circles

**Q65. Find the next term: A1Z, C3X, E5V, G7T, ?**

(a) I9R (b) H9R
(c) I9S (d) J9R

**Q66. A cube painted yellow on all faces is cut into 27 small cubes. How many small cubes have NO face painted?**

(a) 1 (b) 8
(c) 6 (d) 0

**Q67. Find the odd pair out:**

(a) 11 - 121 (b) 12 - 144
(c) 15 - 225 (d) 13 - 170

**Q68. If PEN is coded as 16514, how is BOOK coded?**

(a) 2151511 (b) 2151152
(c) 2141411 (d) 2151512

**Q69. If 1st January 2024 was Monday, what day was 1st January 2025? (Note: 2024 is a leap year)**

a) Wednesday c) Thursday

b) Tuesday d) Friday

**Q70. What is the angle between the hands of a clock at 6:00?**

(a) 180 degrees (b) 90 degrees
(c) 120 degrees (d) 360 degrees

**Q71. Select the mirror image of '1509' when mirror is placed at the bottom (Water image):**

(a) Inverted digits vertically (b) Reversed horizontally
(c) Same number (d) None

**Q72. How many squares are there in a standard 3x3 chessboard grid?**

(a) 14 (b) 9
(c) 10 (d) 16

**Q73. Find missing number: Matrix [ [3, 4, 25], [5, 12, 169], [6, 8, ?] ]**

(a) 100 (b) 80
(c) 90 (d) 120

**Q74. Choose the word least like others:**

(a) Jan (b) March
(c) May (d) April

**Q75. If A is the brother of B, B is the sister of C, and C is the father of D, how is A related to D?**

(a) Uncle (b) Father
(c) Grandfather (d) Brother

## SECTION C: ENGLISH LANGUAGE

**Q76. Read the passage: 'Water conservation is the practice of using water efficiently to reduce unnecessary water usage. Fresh water is a limited resource, and saving it ensures its availability for future generations.' — What is water conservation?**

(a) Using water efficiently (b) Wasting water
(c) Storing salt water (d) Polluting rivers

**Q77. Why is fresh water considered critical according to the passage?**

(a) It is a limited resource (b) It is infinite
(c) It is toxic (d) It cannot be saved

**Q78. What is the main benefit of conserving water?**

(a) Ensures availability for future generations (b) Increases water bills
(c) Dries up rivers (d) Causes floods

**Q79. Choose a suitable title for the passage:**

(a) Importance of Water Conservation (b) Water Pollution
(c) Oceanic Life (d) Rainfall Patterns

**Q80. Identify the correctly spelt word:**

(a) Grammar (b) Grammer
(c) Grammar (d) Gramer

**Q81. Choose the synonym of 'CANDID':**

(a) Frank (b) Secretive
(c) Dishonest (d) Shy

**Q82. Choose the antonym of 'ANCIENT':**

(a) Modern (b) Old
(c) Historical (d) Antique

**Q83. Fill in the blank: He is senior _____ me in service.**

(a) to (b) than
(c) from (d) with

**Q84. Change to Passive Voice: 'Who wrote the Ramayana?'**

(a) By whom was the Ramayana written? (b) Who was written Ramayana?
(c) By whom Ramayana written? (d) Whom wrote the Ramayana?

**Q85. Change to Indirect Speech: She said to me, 'Where are you going?'**

(a) She asked me where I was going. (b) She asked me where was I going.
(c) She told me where I am going. (d) She asked me that where I went.

**Q86. Fill in the blank with suitable modal: _____ I come in, Sir?**

(a) May (b) Can
(c) Might (d) Would

**Q87. Identify the clause: 'Unless you work hard, you cannot pass.'**

(a) Adverb Clause of Condition (b) Adjective Clause
(c) Noun Clause (d) Relative Clause

**Q88. Fill in the blank: Bread and butter _____ his wholesome breakfast.**

(a) is (b) are
(c) were (d) have

**Q89. Complete the question tag: He never tells a lie, _____?**

(a) does he (b) doesn't he
(c) isn't he (d) did he

**Q90. Choose the meaning of idiom: 'A blessing in disguise'**

(a) A good outcome from a bad situation (b) A hidden curse
(c) A religious blessing (d) A false statement

**Q91. Fill in the blank: She is _____ M.A. in English.**

(a) an (b) a
(c) the (d) no article

**Q92. Identify the grammatically correct sentence:**

(a) Neither of the two candidates was selected. (b) Neither of the two candidates were selected.
(c) Neither of candidates have selected. (d) Neither candidates was selected.

**Q93. One-word substitution for: 'A person who does not believe in God'**

(a) Atheist (b) Theist
(c) Agnostic (d) Devotee

**Q94. Choose the antonym of 'CONCORD':**

(a) Discord (b) Agreement
(c) Harmony (d) Peace

**Q95. Fill in the blank: If I _____ a king, I would help the poor.**

(a) were (b) was
(c) am (d) had been

**Q96. Rearrange jumbled words: P: necessary / Q: health / R: exercise / S: is for good**

(a) R-S-P-Q / Exercise is necessary for good health (b) P-Q-R-S
(c) Q-R-S-P (d) S-P-Q-R

**Q97. Fill in the blank: He worked hard _____ he might win the prize.**

(a) so that (b) because
(c) lest (d) unless

**Q98. Identify the collective noun in: 'A herd of cattle was grazing in the field.'**

(a) Herd (b) Cattle
(c) Grazing (d) Field

**Q99. Fill in the blank: The news _____ true.**

(a) is (b) are
(c) were (d) have

**Q100. One-word substitution for: 'A life history of a person written by himself'**

(a) Autobiography (b) Biography
(c) Memoir (d) History

## SECTION D: GENERAL SCIENCE

**Q101. Which agricultural implement is used for removing weeds and loosening the soil?**

(a) Hoe (b) Combine
(c) Seed drill (d) Sickle

**Q102. Which non-metal is stored in water to prevent it from catching fire in air?**

(a) Phosphorus (b) Sodium
(c) Sulphur (d) Carbon

**Q103. Polycot is obtained by blending which two fibres?**

(a) Polyester and Cotton (b) Polyester and Wool
(c) Polymer and Cotton (d) Rayon and Cotton

**Q104. Which metal is stored under kerosene oil due to its extreme reactivity with air and water?**

(a) Sodium (b) Iron
(c) Copper (d) Gold

**Q105. Which oxide is basic in nature?**

(a) Magnesium oxide (b) Sulphur dioxide
(c) Carbon dioxide (d) Nitrogen dioxide

**Q106. Incomplete combustion of carbon fuels releases which poisonous, colourless gas?**

(a) Carbon monoxide (b) Carbon dioxide
(c) Methane (d) Sulphur dioxide

**Q107. The zone of a candle flame which is the hottest and undergoes complete combustion is:**

(a) Outer non-luminous zone (b) Middle luminous zone
(c) Inner dark zone (d) Blue base zone

**Q108. Red Data Book maintains an international record of:**

(a) Endangered species (b) Extinct species
(c) Forest cover (d) National parks

**Q109. The fluid content inside the cell membrane excluding the nucleus is called:**

(a) Cytoplasm (b) Protoplasm
(c) Nucleoplasm (d) Vacuole

**Q110. Which organelle is present ONLY in plant cells and responsible for photosynthesis?**

(a) Chloroplast (b) Centrosome
(c) Lysosome (d) Mitochondria

**Q111. External fertilization takes place in which of the following animals?**

(a) Frog (b) Hen
(c) Human (d) Cow

**Q112. Metamorphosis in tadpoles to transform into adult frogs is controlled by which hormone?**

(a) Thyroxin (b) Insulin
(c) Adrenaline (d) Growth hormone

**Q113. The pressure exerted by a liquid at the bottom of a container depends on the:**

(a) Height of the liquid column (b) Shape of the container
(c) Base area of container (d) Colour of liquid

**Q114. Which force is responsible for holding the planets in orbit around the Sun?**

(a) Gravitational force (b) Electrostatic force
(c) Magnetic force (d) Frictional force

**Q115. Which type of friction is the smallest among static, sliding, and rolling friction?**

(a) Rolling friction (b) Static friction
(c) Sliding friction (d) Fluid friction

**Q116. Sound cannot travel through which of the following mediums?**

(a) Vacuum (b) Water
(c) Air (d) Steel

**Q117. The pitch or shrillness of a sound depends on its:**

(a) Frequency (b) Amplitude
(c) Loudness (d) Speed

**Q118. Electroplating a layer of Zinc over Iron to protect it from rusting is called:**

(a) Galvanisation (b) Anodising
(c) Alloying (d) Vulcanisation

**Q119. The instrument used to detect and measure electric charge on a body is:**

(a) Electroscope (b) Ammeter
(c) Voltmeter (d) Galvanometer

**Q120. The tectonic plates of Earth's crust slide past each other along fault lines causing:**

(a) Earthquakes (b) Tsunamis
(c) Cyclones (d) Volcanic eruptions

**Q121. Dispersion of white light into its seven constituent colors occurs through a:**

(a) Glass prism (b) Plane mirror
(c) Concave lens (d) Glass block

**Q122. The human eye lens forms which type of image on the retina?**

(a) Real and inverted (b) Virtual and erect
(c) Real and erect (d) Virtual and inverted

**Q123. Which planet is the largest planet in our solar system?**

(a) Jupiter (b) Saturn
(c) Neptune (d) Uranus

**Q124. Constellation consisting of seven prominent stars also known as 'Saptarshi' is:**

(a) Ursa Major (Great Bear) (b) Orion
(c) Cassiopeia (d) Scorpius

**Q125. Which atmospheric layer absorbs harmful ultraviolet (UV) radiation from the Sun?**

(a) Ozone layer (b) Troposphere
(c) Ionosphere (d) Exosphere

## SECTION E: SOCIAL SCIENCE

**Q126. The Ryotwari system of land revenue collection was introduced in South India by:**

(a) Thomas Munro and Alexander Read (b) Lord Cornwallis
(c) Holt Mackenzie (d) Warren Hastings

**Q127. Who led the armed tribal rebellion against British rule in Chota Nagpur region?**

(a) Birsa Munda (b) Sidhu and Kanhu
(c) Titu Mir (d) Alluri Sitarama Raju

**Q128. Where was the main center of the Revolt of 1857 under the leadership of Begum Hazrat Mahal?**

(a) Lucknow (b) Jhansi
(c) Kanpur (d) Delhi

**Q129. The Vernacular Press Act was enacted by the British Government in 1878 to:**

(a) Curb Indian language newspapers criticizing British policies (b) Promote English literacy
(c) Support native journalists (d) Establish press freedom

**Q130. The Indian National Congress (INC) was established in December 1885 at:**

(a) Bombay (b) Calcutta
(c) Madras (d) Delhi

**Q131. The Partition of Bengal in 1905 was executed by which Viceroy?**

(a) Lord Curzon (b) Lord Ripon
(c) Lord Minto (d) Lord Hardinge

**Q132. Non-Cooperation Movement was called off by Mahatma Gandhi in February 1922 due to the violent incident at:**

(a) Chauri Chaura (b) Jallianwala Bagh
(c) Kakori (d) Dandi

**Q133. Resources whose total quantity is known and are currently being used are called:**

(a) Actual resources (b) Potential resources
(c) Abiotic resources (d) Ubiquitous resources

**Q134. Breaker zones, contour ploughing, and shelterbelts are methods used for:**

(a) Soil conservation (b) Water purification
(c) Mining minerals (d) Urban planning

**Q135. Which crop requires high temperature, heavy rainfall, high humidity, and standing water?**

(a) Paddy (Rice) (b) Wheat
(c) Maize (d) Cotton

**Q136. The first successful modern textile mill in India was established in 1854 at:**

(a) Mumbai (Bombay) (b) Ahmedabad
(c) Kolkata (d) Coimbatore

**Q137. Which state in India has the highest population according to Census 2011?**

(a) Uttar Pradesh (b) Maharashtra
(c) Bihar (d) West Bengal

**Q138. The Constitution of India was adopted by the Constituent Assembly on:**

(a) 26 November 1949 (b) 26 January 1950
(c) 15 August 1947 (d) 30 January 1948

**Q139. Indian Secularism implies that the Indian State:**

(a) Does not promote or favor any single religion (b) Promotes the majority religion
(c) Bans all religious practices (d) Is controlled by religious leaders

**Q140. How many members are nominated by the President of India to the Rajya Sabha?**

(a) 12 members (b) 2 members
(c) 10 members (d) 15 members

**Q141. The highest judicial court of appeal in an Indian state is the:**

(a) High Court (b) District Court
(c) Supreme Court (d) Sessions Court

**Q142. The term 'Marginalisation' refers to:**

(a) Forcing a community to the fringes of social, economic, and political life (b) Providing equal privileges to all
(c) Promoting economic growth (d) Erecting physical borders

**Q143. Which Act safeguards Scheduled Castes and Scheduled Tribes against atrocities and discrimination?**

(a) POA Act 1989 (Prevention of Atrocities) (b) Dowry Prohibition Act
(c) RTI Act 2005 (d) Factories Act

**Q144. Golden Fibre refers to which natural commercial crop?**

(a) Jute (b) Cotton
(c) Silk (d) Hemp

**Q145. Bhilai, Rourkela, Durgapur, and Bokaro are famous centers for which major industry?**

(a) Iron and Steel Industry (b) Automobile Industry
(c) Pharmaceuticals (d) Cotton Textile

**Q146. Who established the Satyashodhak Samaj in 1873 to fight caste oppression?**

(a) Jyotirao Phule (b) Raja Rammohan Roy
(c) Swami Vivekananda (d) Keshab Chandra Sen

**Q147. Under Article 21 of the Indian Constitution, the Right to Life includes the Right to:**

(a) Clean Water and Pollution-free Air (b) Free Internet
(c) Free Luxury Housing (d) Foreign Travel for all

**Q148. Which fundamental right allows citizens to move court if their rights are violated?**

(a) Right to Constitutional Remedies (Article 32) (b) Right to Equality
(c) Right to Freedom of Religion (d) Right against Exploitation

**Q149. Who was the last Governor-General and first Viceroy of India under British Crown rule in 1858?**

(a) Lord Canning (b) Lord Dalhousie
(c) Lord Ripon (d) Lord Curzon

**Q150. The age group considered as 'working population' in demographic studies is:**

(a) 15 to 59 years (b) 0 to 14 years
(c) 60 to 80 years (d) 18 to 70 years

## PART B: MASTER ANSWER KEY

| Q.No | Ans | Q.No | Ans | Q.No | Ans | Q.No | Ans | Q.No | Ans |
|---|---|---|---|---|---|---|---|---|---|
| Q1 | a | Q2 | a | Q3 | a | Q4 | a | Q5 | a |
| Q6 | a | Q7 | a | Q8 | a | Q9 | a | Q10 | a |
| Q11 | a | Q12 | a | Q13 | a | Q14 | a | Q15 | a |
| Q16 | a | Q17 | a | Q18 | a | Q19 | a | Q20 | a |
| Q21 | a | Q22 | a | Q23 | a | Q24 | a | Q25 | a |
| Q26 | a | Q27 | a | Q28 | a | Q29 | a | Q30 | a |
| Q31 | a | Q32 | a | Q33 | a | Q34 | a | Q35 | a |
| Q36 | a | Q37 | a | Q38 | a | Q39 | a | Q40 | a |
| Q41 | a | Q42 | a | Q43 | a | Q44 | a | Q45 | a |
| Q46 | a | Q47 | a | Q48 | a | Q49 | a | Q50 | a |
| Q51 | a | Q52 | a | Q53 | b | Q54 | a | Q55 | a |
| Q56 | a | Q57 | a | Q58 | a | Q59 | a | Q60 | a |
| Q61 | a | Q62 | a | Q63 | a | Q64 | a | Q65 | a |
| Q66 | a | Q67 | d | Q68 | a | Q69 | a | Q70 | a |
| Q71 | a | Q72 | a | Q73 | a | Q74 | d | Q75 | a |
| Q76 | a | Q77 | a | Q78 | a | Q79 | a | Q80 | a |
| Q81 | a | Q82 | a | Q83 | a | Q84 | a | Q85 | a |
| Q86 | a | Q87 | a | Q88 | a | Q89 | a | Q90 | a |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Q91 | a | Q92 | a | Q93 | a | Q94 | a | Q95 | a |
| Q96 | a | Q97 | a | Q98 | a | Q99 | a | Q100 | a |
| Q101 | a | Q102 | a | Q103 | a | Q104 | a | Q105 | a |
| Q106 | a | Q107 | a | Q108 | a | Q109 | a | Q110 | a |
| Q111 | a | Q112 | a | Q113 | a | Q114 | a | Q115 | a |
| Q116 | a | Q117 | a | Q118 | a | Q119 | a | Q120 | a |
| Q121 | a | Q122 | a | Q123 | a | Q124 | a | Q125 | a |
| Q126 | a | Q127 | a | Q128 | a | Q129 | a | Q130 | a |
| Q131 | a | Q132 | a | Q133 | a | Q134 | a | Q135 | a |
| Q136 | a | Q137 | a | Q138 | a | Q139 | a | Q140 | a |
| Q141 | a | Q142 | a | Q143 | a | Q144 | a | Q145 | a |
| Q146 | a | Q147 | a | Q148 | a | Q149 | a | Q150 | a |

# PART C: DETAILED SOLUTIONS

## SECTION A: MATHEMATICS — SOLUTIONS

**Q1. Correct Option: (a)**
Let the other number be x. (-4/3) * x = -16/9 => x = (-16/9) / (-4/3) = (-16/9) * (-3/4) = 4/3.

**Q2. Correct Option: (a)**
2(y - 8) = 3(y - 3) => 2y - 16 = 3y - 9 => y = -7.

**Q3. Correct Option: (a)**
Let tens digit = x, units = y. x + y = 11. Number = 10x + y. (10x + y) + 27 = 10y + x => 9y - 9x = 27 => y - x = 3. Adding x + y = 11 and y - x = 3 gives 2y = 14 => y = 7, x = 4. Number is 47.

**Q4. Correct Option: (a)**
Exterior angle of regular polygon = 360 / n = 360 / 15 = 24 degrees.

**Q5. Correct Option: (a)**
Perimeter = 2(4x + 3x) = 56 => 14x = 56 => x = 4. Sides are 16 cm and 12 cm. Diagonal = sqrt(16^2 + 12^2) = sqrt(256 + 144) = sqrt(400) = 20 cm.

**Q6. Correct Option: (a)**
Number of red queens (Hearts and Diamonds) = 2. Total cards = 52. Probability = 2 / 52 = 1 / 26.

**Q7. Correct Option: (a)**
36^2 = 1296. 1300 - 1296 = 4. Subtracting 4 gives 1296, which is a perfect square.

**Q8. Correct Option: (a)**
59049 = 243^2. Since there are 8 decimal places, square root has 4 decimal places => 0.0243.

**Q9. Correct Option: (a)**
68600 = 2^3 * 7^3 * 5^2. Multiplying by 5 gives 5^3, making it a perfect cube. Smallest multiplier = 5.

**Q10. Correct Option: (a)**
14^3 = 2744. Thus, root_3(-2744) = -14.

**Q11. Correct Option: (a)**
Let original price = x. x + 0.18x = 1239 => 1.18x = 1239 => x = 1239 / 1.18 = Rs 1050.

**Q12. Correct Option: (a)**
A = 12600 * (1 + 10/100)^2 = 12600 * (121 / 100) = 15246. CI = 15246 - 12600 = Rs 2646.

**Q13. Correct Option: (a)**
(5xy - 2yz - 2zx + 10xyz) - (3xy + 5yz - 7zx) = (5-3)xy + (-2-5)yz + (-2+7)zx + 10xyz = 2xy - 7yz + 5zx + 10xyz.

**Q14. Correct Option: (a)**
(100 + 2)^2 = 10000 + 2(100)(2) + 4 = 10404.

**Q15. Correct Option: (a)**
Euler's formula: F + V - E = 2 => 6 + V - 12 = 2 => V - 6 = 2 => V = 8.

**Q16. Correct Option: (a)**
Area = (1/2) * (24 + 20) * 15 = (1/2) * 44 * 15 = 22 * 15 = 330 cm^2.

**Q17. Correct Option: (a)**
Volume = pi * r^2 * h => 1540 = (22/7) * r^2 * 10 => 154 = (22/7) * r^2 => r^2 = (154 * 7)/22 = 49 => r = 7 cm.

**Q18. Correct Option: (a)**
6 * a^2 = 486 => a^2 = 81 => a = 9 cm. Volume = a^3 = 9^3 = 729 cm^3.

**Q19. Correct Option: (a)**
Simplify powers of 2, 3, 5: 3^-5 * 2^-5 * 5^-5 * 5^3 / (5^-7 * 2^-5 * 3^-5) = 5^(-5 + 3 - (-7)) = 5^5 = 3125.

**Q20. Correct Option: (a)**
Inverse proportion: 15 * 24 = x * 18 => x = 360 / 18 = 20 men.

**Q21. Correct Option: (a)**
16x^2 - 25y^2 = (4x)^2 - (5y)^2 = (4x - 5y)(4x + 5y).

**Q22. Correct Option: (a)**
p^2 + 6p + 8 = p^2 + 2p + 4p + 8 = (p + 2)(p + 4).

**Q23. Correct Option: (a)**
Sum of digits = 3 + 1 + z + 5 = 9 + z. For multiple of 3, z can be 0, 3, 6, 9. Sum = 0 + 3 + 6 + 9 = 18.

**Q24. Correct Option: (a)**
Distance along x-axis = |7 - 2| = 5 units.

**Q25. Correct Option: (a)**
x^((a-b)(a+b)) * x^((b-c)(b+c)) * x^((c-a)(c+a)) = x^(a^2-b^2 + b^2-c^2 + c^2-a^2) = x^0 = 1.

**Q26. Correct Option: (a)**
Perimeter = 2(5x + 3x) = 160 => 16x = 160 => x = 10. Dimensions = 50 m and 30 m. Area = 50 * 30 = 1500 m^2.

**Q27. Correct Option: (a)**
Radius = 3.5 m = 7/2 m. Volume = pi * r^2 * h = (22/7) * (7/2) * (7/2) * 20 = 22 * (7/4) * 20 = 770 m^3.

**Q28. Correct Option: (a)**
Diagonal = sqrt(l^2 + b^2 + h^2) = sqrt(12^2 + 9^2 + 8^2) = sqrt(144 + 81 + 64) = sqrt(289) = 17 m.

**Q29. Correct Option: (a)**
[(3) - (4)]^-1 = [-1]^-1 = -1.

**Q30. Correct Option: (a)**
Speed = 180 / 3 = 60 km/h. Distance in 5 hours = 60 * 5 = 300 km.

**Q31. Correct Option: (a)**
a^4 - b^4 = (a^2 - b^2)(a^2 + b^2) = (a - b)(a + b)(a^2 + b^2).

**Q32. Correct Option: (a)**
28x^4 / 56x = (28/56) * x^(4-1) = (1/2) x^3.

**Q33. Correct Option: (a)**
For multiple of 6, it must be even (x in {0,2,4,6,8}) and sum 2+4+x = 6+x must be divisible by 3. So x can be 0 or 6.

**Q34. Correct Option: (a)**
Class intervals: 5-15, 15-25, 25-35, 35-45, 45-55. Upper limit of 5th interval = 55.

**Q35. Correct Option: (a)**
Sum of 5 numbers = 5 * 27 = 135. Sum of 4 numbers = 4 * 25 = 100. Excluded number = 135 - 100 = 35.

**Q36. Correct Option: (a)**
2^(x - 3) = 2^4 => x - 3 = 4 => x = 7.

**Q37. Correct Option: (a)**
Third angle = 180 - (55 + 65) = 180 - 120 = 60 degrees.

**Q38. Correct Option: (a)**
Half-diagonals are 8 cm and 6 cm. Side = sqrt(8^2 + 6^2) = sqrt(64 + 36) = 10 cm. Perimeter = 4 * 10 = 40 cm.

**Q39. Correct Option: (a)**
Volume = (2/3) * pi * r^3 = (2/3) * (22/7) * (3.5)^3 = (44/21) * 42.875 = 89.83 cm^3.

**Q40. Correct Option: (a)**
5^6 * 5^-4 = 5^2 = 25.

**Q41. Correct Option: (a)**
Revolutions per sec = 360 / 60 = 6 rev/s. Radians per sec = 6 * 2 pi = 12 pi radians.

**Q42. Correct Option: (a)**
100 - 9x^2 = (10)^2 - (3x)^2 = (10 - 3x)(10 + 3x).

**Q43. Correct Option: (a)**
Numerator = (z - 6)(z + 2). Dividing by (z + 2) gives (z - 6).

**Q44. Correct Option: (a)**
Number of outcomes = 2^3 = 8.

**Q45. Correct Option: (a)**
23 * 23 = 529. So sqrt(529) = 23.

**Q46. Correct Option: (a)**
1250 = 2 * 625 = 2 * (25)^2. Dividing by 2 leaves 625 = 25^2. Smallest divisor = 2.

**Q47. Correct Option: (a)**
Let CP = 100. MP = 120. SP = 120 - 10% of 120 = 108. Profit = 8%.

**Q48. Correct Option: (a)**
4.5 * 10^-4 = 0.00045.

**Q49. Correct Option: (a)**
Inverse proportion: 10 * 8 = 16 * D => D = 80 / 16 = 5 days.

**Q50. Correct Option: (a)**
Sum of interior angles = (n - 2) * 180 degrees.

## SECTION B: INTELLIGENCE — SOLUTIONS

**Q51. Correct Option: (a)**
Each letter is shifted +1 step forward. S->T, C->D, H->I, O->P, O->P, L->M. SCHOOL -> TDIPPM.

**Q52. Correct Option: (a)**
Pattern: (x * 2) + 1. 63 * 2 + 1 = 127.

**Q53. Correct Option: (b)**
Cube is a 3D solid shape, whereas Square, Rectangle, and Triangle are 2D plane figures.

**Q54. Correct Option: (a)**
'Her mother's only daughter' is the lady herself. So she is his wife.

**Q55. Correct Option: (a)**
Replacing: 15 + 5 * 2 / 4 - 6 = 15 + (10 / 4) - 6 = 15 + 2.5 - 6 = 11.5.

**Q56. Correct Option: (a)**
Eyes are used for seeing; ears are used for hearing.

**Q57. Correct Option: (a)**
Walking West 10m, Left (South) 10m, Left (East) 10m puts him 10m South of the starting point.

**Q58. Correct Option: (a)**
Order: Seed (1) -> Plant (2) -> Tree (4) -> Flower (5) -> Fruit (3).

**Q59. Correct Option: (a)**
Position from right = 30 - 10 + 1 = 21st.

**Q60. Correct Option: (a)**
Steps: +2, +3, +4, +5, +6. P (16) + 6 = 22 (V).

**Q61. Correct Option: (a)**
The word 'DANGER' contains 'A' and 'R', which are not present in 'KNOWLEDGE'.

**Q62. Correct Option: (a)**
8^2 = 64; 11^2 = 121.

**Q63. Correct Option: (a)**
S(19) + U(21) + N(14) = 54. M(13) + O(15) + O(15) + N(14) = 57.

**Q64. Correct Option: (a)**
Delhi is inside India, and India is inside Asia (three concentric circles).

**Q65. Correct Option: (a)**
First letter +2 (A, C, E, G -> I). Number +2 (1, 3, 5, 7 -> 9). Last letter -2 (Z, X, V, T -> R). Next = I9R.

**Q66. Correct Option: (a)**
Cubes with 0 faces painted = (n - 2)^3 = (3 - 2)^3 = 1^3 = 1.

**Q67. Correct Option: (d)**
In (13 - 170), 13^2 = 169, not 170.

**Q68. Correct Option: (a)**
P=16, E=5, N=14. B=2, O=15, O=15, K=11 => 2151511.

**Q69. Correct Option:**

**Q70. Correct Option: (a)**
At 6 o'clock, the hands form a straight line of 180 degrees.

**Q71. Correct Option: (a)**
Mirror at bottom produces vertical inversion (water image).

**Q72. Correct Option: (a)**
Number of squares = 1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14.

**Q73. Correct Option: (a)**
Pattern: a^2 + b^2 = c. 3^2 + 4^2 = 25. 5^2 + 12^2 = 169. 6^2 + 8^2 = 36 + 64 = 100.

**Q74. Correct Option: (d)**
Jan, March, and May have 31 days, whereas April has 30 days.

**Q75. Correct Option: (a)**
A is brother of C (father of D). Thus A is the paternal uncle of D.

## SECTION C: ENGLISH LANGUAGE — SOLUTIONS

**Q76. Correct Option: (a)**
The passage defines water conservation as using water efficiently.

**Q77. Correct Option: (a)**
The passage states that fresh water is a limited resource.

**Q78. Correct Option: (a)**
Saving water ensures its availability for future generations.

**Q79. Correct Option: (a)**
'Importance of Water Conservation' summarizes the passage core topic.

**Q80. Correct Option: (a)**
'Grammar' is spelt with 'ar' at the end.

**Q81. Correct Option: (a)**
Candid means truthful and straightforward; frank.

**Q82. Correct Option: (a)**
Modern is the direct antonym of ancient.

**Q83. Correct Option: (a)**
Adjectives ending in '-ior' (senior, junior, superior) take the preposition 'to'.

**Q84. Correct Option: (a)**
'Who' changes to 'By whom' and past tense verb 'wrote' changes to 'was written' in passive voice.

**Q85. Correct Option: (a)**
Reporting verb 'said to' changes to 'asked', interrogative structure becomes assertive ('I was going').

**Q86. Correct Option: (a)**
'May' is used for seeking formal permission.

**Q87. Correct Option: (a)**
'Unless you work hard' specifies a condition under which the main clause occurs.

**Q88. Correct Option: (a)**
When two nouns express a single combined idea ('bread and butter'), they take a singular verb 'is'.

**Q89. Correct Option: (a)**
Since 'never' makes the sentence negative, it takes a positive question tag 'does he?'.

**Q90. Correct Option: (a)**
'A blessing in disguise' refers to something that seems bad at first but results in something good.

**Q91. Correct Option: (a)**
'M.A.' starts with a vowel sound /em/, taking the indefinite article 'an'.

**Q92. Correct Option: (a)**
'Neither of' referring to two items takes a singular verb 'was'.

**Q93. Correct Option: (a)**
An atheist is a person who disbelieves or lacks belief in the existence of God.

**Q94. Correct Option: (a)**
Discord (conflict) is the antonym of concord (harmony).

**Q95. Correct Option: (a)**
Subjunctive mood for hypothetical/unreal conditions uses 'were' with all subjects.

**Q96. Correct Option: (a)**
Proper order: Exercise is necessary for good health.

**Q97. Correct Option: (a)**
'So that' expresses purpose followed by 'might' or 'may'.

**Q98. Correct Option: (a)**
'Herd' is a collective noun describing a group of animals.

**Q99. Correct Option: (a)**
'News' is an uncountable singular noun taking 'is'.

**Q100. Correct Option: (a)**
An autobiography is an account of a person's life written by that person.

## SECTION D: GENERAL SCIENCE — SOLUTIONS

**Q101. Correct Option: (a)**
A hoe is a traditional tool used for pulling out weeds and loosening topsoil.

**Q102. Correct Option: (a)**
White phosphorus reacts vigorously with atmospheric oxygen and is kept immersed in water.

**Q103. Correct Option: (a)**
Polycot is a blended fabric produced by combining polyester and cotton.

**Q104. Correct Option: (a)**
Sodium and Potassium react violently with air/moisture and are stored under kerosene.

**Q105. Correct Option: (a)**
Metallic oxides like Magnesium oxide are basic, whereas non-metallic oxides are acidic.

**Q106. Correct Option: (a)**
Incomplete combustion yields Carbon monoxide (CO), which reduces oxygen-carrying capacity of blood.

**Q107. Correct Option: (a)**
The outermost non-luminous zone has plenty of oxygen supply and is the hottest region.

**Q108. Correct Option: (a)**
Red Data Book published by IUCN keeps a record of all endangered plants and animals.

**Q109. Correct Option: (a)**
Cytoplasm is the jelly-like substance surrounding organelles outside the nucleus.

**Q110. Correct Option: (a)**
Chloroplasts contain chlorophyll required for plant photosynthesis.

**Q111. Correct Option: (a)**
Frogs and fishes release eggs and sperms into water where external fertilization occurs.

**Q112. Correct Option: (a)**
Thyroxin produced by the thyroid gland (requiring iodine) controls tadpole metamorphosis.

**Q113. Correct Option: (a)**
Liquid pressure increases directly with the height/depth of the liquid column (P = h * rho * g).

**Q114. Correct Option: (a)**
Solar gravitational attraction supplies the centripetal force keeping planets in orbit.

**Q115. Correct Option: (a)**
Rolling friction is significantly smaller than sliding and static friction.

**Q116. Correct Option: (a)**
Sound requires a material medium for propagation and cannot travel through vacuum.

**Q117. Correct Option: (a)**
Higher vibration frequency produces a higher pitch/shrill sound.

**Q118. Correct Option: (a)**
Galvanisation coats iron with a thin zinc layer to prevent oxidation.

**Q119. Correct Option: (a)**
An electroscope detects the presence and nature of electric charges.

**Q120. Correct Option: (a)**
Sudden shifting or slipping of tectonic plates along faults triggers seismic shockwaves (earthquakes).

**Q121. Correct Option: (a)**
A glass prism refracts different wavelengths at different angles, splitting white light into a spectrum.

**Q122. Correct Option: (a)**
The convex eye lens forms a real, inverted image on the retina which the brain perceives erect.

**Q123. Correct Option: (a)**
Jupiter is the largest planet in the solar system.

**Q124. Correct Option: (a)**
Saptarshi is part of the Ursa Major constellation.

**Q125. Correct Option: (a)**
The ozone layer in the stratosphere shields Earth by absorbing solar UV rays.

## SECTION E: SOCIAL SCIENCE — SOLUTIONS

**Q126. Correct Option: (a)**
Thomas Munro introduced the Ryotwari settlement directly with ryots (cultivators) in the Madras region.

**Q127. Correct Option: (a)**
Birsa Munda led the Munda Ulgulan rebellion demanding tribal freedom and land rights.

**Q128. Correct Option: (a)**
Begum Hazrat Mahal led the revolt at Lucknow, Awadh.

**Q129. Correct Option: (a)**
The Vernacular Press Act silenced Indian language newspapers voicing anti-British dissent.

**Q130. Correct Option: (a)**
The INC held its first session in Bombay under the presidency of WC Bonnerjee.

**Q131. Correct Option: (a)**
Lord Curzon partitioned Bengal in 1905, triggering the widespread Swadeshi Movement.

**Q132. Correct Option: (a)**
Gandhi suspended the movement after protesters set fire to a police station at Chauri Chaura.

**Q133. Correct Option: (a)**
Actual resources are surveyed, quantified, and actively utilized.

**Q134. Correct Option: (a)**
Contour ploughing and shelterbelts prevent soil erosion by wind and water.

**Q135. Correct Option: (a)**
Rice is a Kharif tropical crop needing rich alluvial soil and abundant water.

**Q136. Correct Option: (a)**
The first modern cotton textile mill was set up in Bombay in 1854.

**Q137. Correct Option: (a)**
Uttar Pradesh is India's most populous state.

**Q138. Correct Option: (a)**
The Constitution was adopted on 26 November 1949 and came into force on 26 January 1950.

**Q139. Correct Option: (a)**
Indian secularism maintains principled distance from all religions without endorsing any single faith.

**Q140. Correct Option: (a)**
The President nominates 12 members with distinguished contributions in art, literature, science, and social service.

**Q141. Correct Option: (a)**
The High Court is the head of the judiciary within a State.

**Q142. Correct Option: (a)**
Marginalisation refers to social exclusion and denial of equal opportunities.

**Q143. Correct Option: (a)**
The SC/ST (Prevention of Atrocities) Act was enacted in 1989.

**Q144. Correct Option: (a)**
Jute is called the Golden Fibre due to its silky texture and high export value.

**Q145. Correct Option: (a)**
These public sector centers house India's major integrated iron and steel plants.

**Q146. Correct Option: (a)**
Jyotirao Phule founded Satyashodhak Samaj to champion equal rights for lower castes.

**Q147. Correct Option: (a)**
Supreme Court rulings expanded Article 21 to include access to clean water and a healthy environment.

**Q148. Correct Option: (a)**
Article 32 grants the Right to Constitutional Remedies, called the 'heart and soul' of the Constitution.

**Q149. Correct Option: (a)**
Lord Canning became the first Viceroy following the Government of India Act 1858.

**Q150. Correct Option: (a)**
The economically productive working age population spans 15 to 59 years.